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Generation and deployment of honeytokens in relational databases for cyber deception

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Highlights

- We designed a Deception Based Framework for Data Breach Detection.
- Our framework uses the Synthetic Data Vault library to generate the Synthetic Data.
- Generated Synthetic Data will be deployed as Honeypot to detect the data breach and the misuse of stolen data.
- We present the in-depth analysis of the obtained results.
- Discussed the deployment strategies for detection of data breach and monitoring of sensitive records.

Abstract

Despite considerable investments in database security, global statistics indicate an exponential increase in data breaches. Organizations are often unaware of data breaches for weeks, months, or even years. Sufficient for adversaries to compromise and ex-filtrate business or mission-critical data. Recent research suggests using honeytokens for early detection of data breaches in organizations. Existing honeypot generation methods rely on regular expressions, rule mining, constraint satisfaction, or representation learning, which are complex and limited to a few attributes. We created a framework for generating and deploying honeytokens in relational databases that actively monitor sensitive records and quickly detect data breaches and their misuse. To generate the honeypot we have used the hierarchical machine learning algorithm which uses a recursive technique to model the parent-child relationships of multi-table databases. The proposed method enables the organization to take remedial action to reduce the impact of data breaches and complement existing database security solutions.

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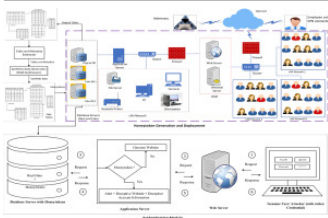
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Introduction

The exponential increase in the use of database services over the Internet has enhanced the reachability of organizations worldwide. In today's digital age, managing and organizing electronically stored data effectively is crucial for the efficient operations of organizations and government entities. Databases play a vital role in the organization of such aspects. According to the structure of data being stored in the database, it can be classified as hierarchical (IBM, 1960, Henry, 1969), relational (Codd, 1970), network (Saxton and Raghavan, 1990), and object-oriented database (Atkinson et al., 1990). The relational database (MySQL, Oracle, SQL Server) consists of numerous related tables such as employee, customer, transaction, product, and services, which are crucial for typical organizations and government agencies. The data and information stored in organizational databases are sensitive and often considered the most valuable assets. Protecting such databases for organizations (small, medium, or large) is paramount. Attacks such as data ex-filtration, identity theft, compromise of credentials, etc., may lead to data loss, loss of privacy, misuse of sensitive data, damage to reputation, customer mistrust, etc. Among all the potential adversaries, detecting malicious insiders (Saxena et al., 2020) is challenging because they are technically skilled, highly motivated, and have access to extensive resources. The best examples are a disgruntled employee who wishes to sell sensitive information to an overseas competitor for personal benefits or a spy working for a foreign country to compromise and endanger national security. Addressing such insider threats is necessary to obtain foolproof security. The attack on databases for government organizations, businesses, and institutions continues to be a critical concern. According to a global report, the average cost of insider threats rose by 31% in the last few years (Saxena et al., 2020). It is one of the significant reasons behind data breaches in healthcare, finance, education, and government organizations. In addition, the global average data breach cost increased by 15% over the last three years (IBM, 2023). The data breaches resulting from compromised credentials take longer to identify, at an average of **327** days (IBM, 2023). So, the active detection of data breach is required to reduce the impact on organization and customers.

Let us take an example of a typical enterprise network of an insurance company as depicted in Fig. 1. It consists of numerous components such as a web server, webmail server, application server, database server, file server, network devices, users (normal, malicious(red)), and adversaries. Among all components, the database server present in the enterprise is crucial for an organization. The database contains personally identifiable information about employees and customers and stores business and mission-critical data. The attacker (external or internal) wants to bypass the existing security controls to steal the critical information stored in the database server. As defenders, we aim to detect data breaches using existing security controls. A plethora of security solutions are present to protect the databases in the enterprise network. Two types of threat actors can threaten the database: external malicious attackers and internally disgruntled employees. The protection mechanisms such as authentication (Chenchev et al., 2021), privilege management (Chadwick and Otenko, 2002, Botha and Eloff, 2001), firewalls (Al-Shaer and Hamed, 2003), intrusion detection and prevention systems (Scarfone et al., 2007), auditing (Natan, 2005), logging (Haerder and Reuter, 1983), classical cryptography (Diffie and Hellman, 1976, Rivest et al., 1978, Koblit, 1987, Miller, 1985) and post-quantum cryptography (Bernstein and Lange, 2017) are standard security solutions effective against external threats; however, not efficient in dealing with insider threats (Maasberg et al., 2020). It occurs when existing employees misuse the access provided by the organization. Due to the unpredictable nature of humans, detecting an attacker's behavior is challenging, and predicting damage due to potential data breaches to the organization is also very difficult. Moreover, the

sophistication of attacker strategies and dependency on third parties for risk identification and implementation of security controls are significant concerns. Despite numerous risks, organizations must grant employees and business partners privileges to function efficiently. A security lapse at a single point (any user or computer in the network, as depicted in Fig. 1) may compromise all the confidential information in the entire database. Since humans are the weakest link, compromising human perception may lead to the failure of all other security controls implemented to protect the database system.

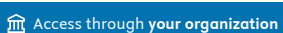
Cyber deception (Galanxhi and Nah, 2007, Han et al., 2018, Ferguson-Walter et al., 2023, Javadpour et al., 2024) is an active defense technique used to mislead the attacker towards an unreal target. In enterprises, it serves multiple purposes to achieve many important goals ranging from understanding attackers' behavior to tricking them into spending resources and time with fake targets. The deception can be implemented at various layers, such as the system (Albanese et al., 2015), application (Izagirre, 2017), network (Spitzner, 2003a, Spitzner, 2003b, Chiang et al., 2018, Qin et al., 2024), and data layers (Spitzner, 2003c, Yuill et al., 2004, Almeshekeh et al., 2015, Dionysiou et al., 2021). Moreover, the cyber deception technique can also be used to secure lightweight-PUF based authentication (Gu et al., 2020). In the age of machine learning and generative AI, there are limited but notable studies where deception has been integrated with adversarial artificial intelligence to safeguard machine learning models (Lopes Antunes and Llopis Sanchez, 2023). These approaches manipulate the decision-making processes of machines by diverting the attacker's focus and misleading their actions. Furthermore, they play a crucial role in predicting and anticipating the behavior of attackers. Dubey et al. (2022) proposed different flavors of side-channel defenses for ML models in the hardware blocks based on Boolean masking techniques. In this work, our focus is on data-level cyber deception, heavily used as a proactive cyber defense to mislead, confuse, or slow down potential attackers. Different types of honeytokens, such as fake login credentials, fake email, fake credit card numbers, etc., are being proposed to know whether the critical resources available in the organization are compromised. Honeytokens are fake digital data artifacts that track down the unauthorized access of sensitive data by purposefully inserting it into a database or password files. We want to create and deploy indistinguishable honeytokens that must mislead the attacker and waste their resource and time. Moreover, the honeytokens must be distributed throughout the organizational network to detect data breaches. Each access to honeytokens will generate an alert to the administrator indicating the database breach or misuse of sensitive data.

Numerous methods have been proposed in the literature to generate honeytokens dealing with fake user accounts, fake emails, fake credit card numbers, fake records, fake tables, or even fake databases. The authors used various methods to generate honeytokens, focusing on character replacement (Juels and Rivest, 2013), rule mining (Cenys et al., 2005, White, 2009), constraint satisfaction (Onaolapo et al., 2016), or representation learning (Bengio et al., 2013) techniques. However, the methods used to create the honeytokens are specific to a particular database; they need manual intervention and are limited to a single table. A typical enterprise database consists of numerous tables related to each other. The generation of fake records for individual tables looks distinguishable, and if adversaries exploit the database, they can easily separate the fake records from the database. To increase the believability of honeytokens, we are generating synthetic data for all the related tables present in a chosen relational database. The defender uses the generated synthetic data as a honeytokens to detect organizational data breaches and the active misuse of sensitive data. There are numerous methods present in the literature to create synthetic records used to complete partial tables, data augmentation, and anonymization in various domains such as healthcare, the US Census, etc. However, there is no work in the literature for detection of data breaches using synthetic data. We have used the HMA synthesizer available in the SDV (Patki et al., 2016) library to generate the synthetic data, which is best suited for generating enterprise databases. The HMA synthesizer works for multi-table synthetic data generation by resembling a database's statistical and structural properties. After successful generation, the administrator deploys the synthetic data (fake database) to detect data breaches in the organization. The alert generation module will monitor the deployed honeytokens and generate an alert to the administrator. To fulfill all the requirements, we have proposed a deception-based framework that takes the database as input and generates honeytokens, their deployment, and monitoring; finally, we evaluated the generated honeytokens using the human evaluation technique.

The organization of the rest of the paper is as follows: Section 2 explores the deception techniques and the use of honeytokens in monitoring and detecting the misuse of sensitive information stored in the database. We will discuss the overall workflow of the proposed method in Section 3 along with the deployment strategies of generated honeytokens. The description of dataset used for our experiment is explained in Section 4. Section 5 deals with the experiments conducted to obtain the realistic honeytokens and their deployment, along with monitoring and generation of alerts. Section 6 will discuss the challenges we faced during the experiment, the limitations of the techniques used to generate honeytokens, and the future scope. Finally, we conclude our work in Section 7.

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Deception has been a survival tactic in Nature and Warfare for millions of years. Living organisms use various deception strategies, including mimicry, camouflage, and nocturnality, as defensive and survival techniques. Deception has likely been a tactic to mislead the enemy in military operations. The Russo-Ukrainian war is the best example of deception in warfare, where Ukrainian soldiers have used the decoy tank to confuse the Russian missile's object detection techniques (The Washington ...

Honeytokens generation and deployment

The overall workflow of the proposed framework is depicted in Fig. 2. The objective of the proposed framework is to generate indistinguishable fake records (honeytokens) for all the tables in the database, followed by effective deployment of honeytokens and their monitoring for active detection of data breaches and misuse of sensitive information. It starts with the extraction module, which extracts the tabular data and their metadata from the database. The output obtained from the extraction ...

Sample database

Due to the lack of availability of real-time relational data, we have used the relational database generated by Fusheng Wang and Carlo Zaniolo at Siemens Corporate Research, publicly available at GitHub (Giuseppe and Patrick, 2023) in '.sql' format. The database contains information about 10,000 employees, their department, and salary details. The database consists of six tables (employee, department, Dept_Emp, Department_Manager, salary and employee_title) connected with foreign keys. We have ...

Experiment and result

Our experiment focuses on the generation of believable synthetic records for financial organizations. Synthetic records can be used as honeytokens to detect data breaches or the misuse of critical information. We implemented the algorithm 1 present in Section 3 to generate the synthetic data. After that, to detect the data breach, we deployed the honeytokens in the database using the strategies mentioned in Section 3.3. To monitor the deployed honeytokens in particular, we have used hash-based ...

Discussion

We focused on the generation of honeytokens for enterprise databases having multiple tables connected with foreign key relationships and their strategic deployment to detect data breaches and misuse of critical information. Since honeytokens are fake data objects, any use in a real-time environment indicates the breach incident by triggering the alert to the administrator. After that, the security

administrator can take appropriate action to reduce the impact of data breaches.
The experimental ...

Conclusion

In this work, we explore the applicability of a hierarchical modeling algorithm (HMA Synthesizer) to generate the honeytokens in a multi-table environment and provide deployment strategies to enhance the existing security controls. The deployed realistic honeytokens can be an effective way for early detection of database breaches and monitoring the misuse of stolen data. Using the deployed honeytokens in real time indicates a possibility of breach, triggering an alert for the administrator. After ...

CRedit authorship contribution statement

Nilin Prabhaker: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ghanshyam S. Bopche:** Validation, Supervision. **Michael Arock:** Supervision. ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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